Zachary Ryan

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EDUCATION

Robert Morris University Pittsburgh, PA
Bachelor of Science in Cybersecurity August 2024 - December 2026
Outstanding Cybersecurity Student Award **[Link]** | Dean's List Recipient GPA: 3.92/4.0

EXPERIENCE

O.F. Mossberg & Sons, Inc. | Artificial Intelligence Intern | Remote August 2026 - Present

- Built and published a Node.js MCP server and Claude Cowork plugin for permission-aware SMB access, encrypted SQLite indexing, and PDF/Office/OCR retrieval; bundled x64/ARM64 runtimes to eliminate manual uploads and simplify setup
- Delivered company-wide AI training for executives and cross-functional teams on safe, responsible AI use and best practices
- Built an AI-assisted workflow to flag outstanding payments across email sources, cutting review from hours to minutes
- Automated Concur credit-card and reimbursement reporting, reducing 5–10 hours of manual work per cycle to minutes

Brixmor Property Group | Cybersecurity Intern | Philadelphia, PA May 2026 - July 2026

- Performed Rapid7 vulnerability management on 25+ high-risk assets, documenting CVEs and remediation recommendations
- Triaged security tickets across a 500+ user enterprise environment using security tools and ticketing platforms
- Assessed 100+ Cisco Umbrella domains, classified DNS risk, and recommended block/allow decisions based on findings
- Assisted with Third-Party Risk Management policy development for vendor onboarding, technology review, security documentation, control assessment, risk acceptance, compliance review, and vendor security evaluation workflows

United States Navy | Machinist's Mate - Nuclear Reactor Department | Norfolk, VA July 2020 - September 2023

- Monitored reactor support systems, logs, gauges, alarms, and indicators to detect abnormal operating conditions
- Diagnosed mechanical and reactor system abnormalities using technical manuals, schematics, and procedures
- Executed incident response procedures under strict safety, documentation, and operational requirements

PROJECTS

ReactorBench-LM | From-Scratch Transformer Research (In Progress) [GitHub]

- Implemented a 15.2M-parameter PyTorch Transformer with causal self-attention, pre-norm blocks, and tied weights
- Trained a 2,048-token SentencePiece BPE tokenizer on synthetic narratives with leakage-resistant dataset splits
- Evaluated 894 held-out examples with parse, schema, and exact-match metrics, exposing gaps despite low validation NLL

Policy-as-Code Docker Network Segmentation Lab | [GitHub]

- Built a Docker lab with public, private, and management zones isolating Flask, PostgreSQL, and attacker containers
- Implemented configurable iptables filtering and JSON policy checks for permitted and blocked inter-container traffic
- Deployed a live demo with isolated Docker sessions, ICMP/TCP probes, and a connectivity dashboard

PUBLICATIONS

Ryan, Z., & Mishra, S. (2026). *Auditable Evidence in AI Cybersecurity Compliance: A Cross-Framework Analysis.* Computers & Security (Submitted June 2026)

- Compared NIST AI RMF, ISO/IEC 42001, EU AI Act, CSA AICM, AWS, and Google Cloud security guidance
- Analyzed evidence gaps across documentation, logging, monitoring, access control, model security, and testing
- Evaluated how logs, access records, monitoring outputs, and assessment artifacts support cybersecurity compliance

Ryan, Z., & Mishra, S. (2026). *Evaluating AI vs Human Phishing Evasion Across Emulated Email Filters and SpamAssassin.* NEDSI Annual Conference (Accepted; presented April 2026) [Publication Link – PDF pp. 189-202]

- Built a 1,260-email dataset comparing AI and human phishing across account, financial, and delivery themes
- Applied Chi-square and Z-tests showing AI phishing evaded selected emulated filters more often than human phishing emails
- Developed a Python filter-emulation framework to compare bypass rates across Gmail, Outlook, Yahoo, and SpamAssassin

SKILLS

Languages & Scripting: Python | SQL | Bash | PowerShell | Java | TypeScript
AI & ML Development: PyTorch | Transformers | Model Training & Evaluation | SentencePiece | Safetensors | MCP
IT & Security Tools: CrowdStrike Falcon | Rapid7 | Azure | Mimecast | Okta | FreshService | Microsoft Defender | Cisco Umbrella
Technology & Development: Linux | Docker | Git/GitHub | Node.js | TCP/IP | Firewalls | Network Segmentation | FastAPI | React

LICENSES AND CERTIFICATIONS

- DoD Secret Clearance | CompTIA Security+ (Scheduled) | Google Cybersecurity Professional Certificate **[Link]**